APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

First Semester B.Tech Degree Examination December 2021.

Course Code : MAT 101.

Course Name : "Linear Algebra and Calculus.

1. Find the rank of the matrix $\begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & -4 \\ 0 & 4 & 0 \end{bmatrix}$

Soln). Let $A = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & -4 \\ 0 & 4 & 0 \end{bmatrix} \underset{\sim}{R_1 \leftrightarrow R_2} \begin{bmatrix} -1 & 0 & -4 \\ 0 & 1 & 0 \\ 0 & 4 & 0 \end{bmatrix}$

$R_3 \to R_3 - 4R_2 \sim \begin{bmatrix} -1 & 0 & -4 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

$\therefore$ Rank of $A = 2$. [2 non zero rows]

2. Find the Eigen values of the matrix $A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$.

Soln) Characteristic equation is

$\lambda^2 - [\text{sum of diagonal elements}]\lambda + |A| = 0$

$\lambda^2 - [1+3]\lambda + 3 = 0 \Rightarrow \lambda^2 - 4\lambda + 3 = 0.$

$\Rightarrow (\lambda-3)(\lambda-1) = 0$

$\lambda_1 = 3 \quad \lambda_2 = 1.$

$\therefore$ Eigen values of A are 3 and 1

Eigen values of A^2 are $3^2, 1^2 = 9, 1$

Eigen values of $A^{-1} = \frac{1}{\lambda_1}, \frac{1}{\lambda_2} = \frac{1}{3}, 1$

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3. If $z = \frac{xy}{x^2+y^2}$ find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$.

$$\frac{\partial z}{\partial x} = \frac{(x^2+y^2)y - (xy)2x}{(x^2+y^2)^2} = \frac{x^2y + y^3 - 2x^2y}{(x^2+y^2)^2}$$

$$= \frac{y^3 - x^2y}{(x^2+y^2)^2}$$

$$\frac{\partial z}{\partial y} = \frac{(x^2+y^2) \cdot x - (xy)2y}{(x^2+y^2)^2} = \frac{x^3 + xy^2 - 2xy^2}{(x^2+y^2)^2}$$

$$= \frac{x^3 - xy^2}{(x^2+y^2)^2}$$

4. Show that the equation $u(x,t) = \sin(x-ct)$, satisfies wave equation $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$

Soln)
$$\frac{\partial u}{\partial t} = \cos(x-ct) \times -c.$$
$$\frac{\partial^2 u}{\partial t^2} = -\sin(x-ct) \times -c \times -c. = -c^2 \sin(x-ct) \quad \text{--- (1)}$$
$$\frac{\partial u}{\partial x} = \cos(x-ct)$$
$$\frac{\partial^2 u}{\partial x^2} = -\sin(x-ct) = \frac{1}{c^2} \frac{\partial^2 u}{\partial t^2} \quad (\text{from (1)})$$
$$\therefore \frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}.$$

5. Evaluate $\int_0^3 \int_0^2 \int_0^1 xyz \, dx\,dy\,dz$.

Soln)
$$\int_0^3 \int_0^2 \left[yz \cdot \frac{x^2}{2}\right]_0^1 dy\,dz = \int_0^3 \int_0^2 \frac{yz}{2} \, dy\,dz.$$
$$= \int_0^3 \frac{z}{2} \left[\frac{y^2}{2}\right]_0^2 dz.$$
$$= \frac{1}{4} \int_0^3 z \cdot 4 \, dz = \int_0^3 z\,dz = \left[\frac{z^2}{2}\right]_0^3 = 9/2$$

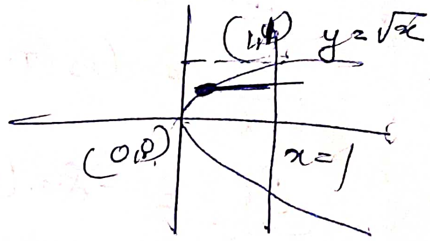
6. Find the mass of the lamina with density $\delta(x,y) = x + 2y$ is bounded by the x axis, the line

$x=1$ and the curve $y^2=x$.

Soln Mass of lamina $= \iint_R \delta(x,y)\, dA$.

$$= \int\int (x+2y)\, \cancel{dy\,dx}\; dx\,dy$$

$$= \int_0^1 \int_{x=y^2}^{x=1} x+2y \; dx\,dy.$$



$$= \int_0^1 \left[\frac{x^2}{2} + 2y\cdot x\right]_{y^2}^{1} dy$$

$$= \int_0^1 \left[\frac{1}{2} + 2y - \left(\frac{y^4}{2} + 2y\cdot y^2\right)\right] dy$$

$$= \int_0^1 \frac{1}{2} + 2y - \frac{y^4}{2} - 2y^3 \; dy$$

$$= \left[\frac{1}{2}y + \frac{2\cdot y^2}{2} - \frac{1}{2}\cdot\frac{y^5}{5} - \frac{2\cdot y^4}{4}\right]_0^1$$

$$= \frac{1}{2} + 1 - \frac{1}{10} - \frac{1}{2}$$

$$= 1 - \frac{1}{10} = 9/10$$

7 Find the rational number represented by the repeating decimal 5.373737....

Soln) $5.3737\ldots = 5 + .373737\ldots$

$= 5 + [0.37 + 0.0037 + 0.000037 + \ldots]$

It is a geometric series. Here $a = 0.37$.

$r = \frac{0.0037}{0.37} = .01$

Series cgs

$\therefore$ Sum, $S = \frac{a}{1-r} = \frac{0.37}{1-0.01} = \frac{0.37}{0.99} = \frac{37}{99}$

$\therefore 5.3737\ldots = 5 + \frac{37}{99} = \frac{5\times 99 + 37}{99}$

$= \frac{532}{99}$

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8. Examine the convergence of $\sum_{k=1}^{\infty} \frac{k^2}{2k^2+3}$.

Soln $\sum_k u_k = \frac{k^2}{2k^2+3} = \frac{k^2}{k^2(2+3/k^2)}$

$= \frac{1}{2+3/k^2}$.

$\lim_{k\to\infty} u_k = \lim_{k\to\infty} \frac{k^2}{2k^2+3} = \lim_{k\to\infty} \frac{1}{2+\frac{3}{k^2}}$.

$= \frac{1}{2} \neq 0$, By divergence test series dgs.

9. Find the Taylor series expansion of $f(x) = \sin \pi x$ about $x = 1/2$

Soln) $f(x) = \sin \pi x$, $f'(x) = \pi \cos \pi x$, $f''(x) = -\pi^2 \sin \pi x$

$f'''(x) = -\pi^3 \cos \pi x$.

$f(1/2) = 1$, $f'(1/2) = 0$ $f''(1/2) = -\pi^2$

$f'''(1/2) = 0$.

$$\therefore \sin \pi x = 1 - \frac{(x-0.5)^2}{2!}\pi^2 + \frac{(x-0.5)^4}{4!}\pi^4 + \cdots$$

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10. If $f(x)$ is a periodic function with period 2π . defined in $[-\pi, \pi]$. Write the Euler's formulas a_0, a_n, b_n for $f(x)$.

Soln) $$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx .$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x) \cos nx \,dx .$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x) \sin(nx)\,dx .$$

11 a). Solve the following linear system of equations using Gauss elimination method

$$x + 2y - z = 3.$$
$$3x - y + 2z = 1$$
$$2x - 2y + 3z = 2.$$

Soln).

$$[A\,B] = \begin{bmatrix} 1 & 2 & -1 & 3 \\ 3 & -1 & 2 & 1 \\ 2 & -2 & 3 & 2 \end{bmatrix} \begin{matrix} R_2 \to R_2 - 3R_1 \\ \sim \\ R_3 \to R_3 - 2R_1 \end{matrix} \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & -7 & 5 & -8 \\ 0 & -6 & 5 & -4 \end{bmatrix}$$

$$\begin{matrix} R_3 \to 7R_3 - 6R_2 \\ \sim \end{matrix} \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & -7 & 5 & -8 \\ 0 & 0 & 5 & 20 \end{bmatrix}$$

By back substitution method,

$$5z = 20.$$
$$z = 20/5 = \underline{4}.$$
$$-7y + 5z = -8.$$
$$-7y + 20 = -8$$
$$-7y = -8 - 20 = -28.$$
$$7y = 28$$
$$y = 28/7 = \underline{4}.$$

$$x+2y-z = 3.$$

$$x+2\cdot 4-4 = 3.$$

$$x+8-4 = 3.$$

$$x+4 = 3.$$

$$x = 3-4 = \underline{\underline{-1}}.$$

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b) Find the eigenvalues and eigenvectors of

$$\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$$

Soln) characteristic eqn is $|A-\lambda I| = 0.$

$$\left| \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} \right| = 0.$$

$$\begin{vmatrix} 8-\lambda & -6 & 2 \\ -6 & 7-\lambda & -4 \\ 2 & -4 & 3-\lambda \end{vmatrix} = 0.$$

$$= (8-\lambda)\left((7-\lambda)(3-\lambda)-16\right)+6\left(-6(3-\lambda)+8\right)$$
$$+2\left(24-2(7-\lambda)\right) = 0.$$

$$\Rightarrow (8-\lambda)(7-\lambda)(3-\lambda)-16(8-\lambda)-36(3-\lambda)+48+48-4(7-\lambda) = 0.$$

$$\Rightarrow (8-\lambda)(7-\lambda)(3-\lambda)-128+16\lambda-108+36\lambda+48+48-28+4\lambda=0$$

$$\Rightarrow (8-\lambda)(7-\lambda)(3-\lambda)+56\lambda-168=0$$

$$\Rightarrow (8-\lambda)(7-\lambda)(3-\lambda)+56[\lambda-3]=0$$

$$\Rightarrow (\lambda-3)\left[(\lambda-8)(7-\lambda)+56\right]=0.$$

$$\Rightarrow (\lambda-3)\left[7\lambda-\lambda^2-56+8\lambda+56\right]=0.$$

$$\Rightarrow (\lambda-3)\left[-\lambda^2+15\lambda\right]=0$$

$$\Rightarrow \lambda(\lambda-3)[15-\lambda]=0$$

$$\lambda = 0, 3, 15$$

<u>Eigen vector corresponding $\lambda=0$.</u>

$$(A-0I)x = 0.$$

$$Ax = 0.$$

$$\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0.$$

$\Rightarrow 8x_1 - 6x_2 + 2x_3 = 0.$

$-6x_1 + 7x_2 - 4x_3 = 0$

$2x_1 - 4x_2 + 3x_3 = 0$

Gauss elimination method

$$\Rightarrow \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix} \quad \begin{array}{l} R_2 \to 4R_2 + 3R_1 \\ R_3 \to 4R_3 - R_1 \end{array} \begin{bmatrix} 8 & -6 & 2 \\ 0 & 10 & -10 \\ 0 & -10 & 10 \end{bmatrix}$$

$$R_3 \to R_3 + R_2 \begin{bmatrix} 8 & -6 & 2 \\ 0 & 10 & -10 \\ 0 & 0 & 0 \end{bmatrix} \qquad R(A) = 2 < n.$$

$10x_2 - 10x_3 = 0.$

$x_2 - x_3 = 0$

$x_2 = x_3$

$8x_1 - 6x_2 + 2x_3 = 0.$

$8x_1 - 6x_3 + 2x_3 = 0$

$8x_1 - 4x_3 = 0.$

$2x_1 - x_3 = 0. \Rightarrow x_3 = 2x_1$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_3/2 \\ x_3 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} 1/2 \\ 1 \\ 1 \end{bmatrix}$$

$\therefore$ Eigen vector corresponding to $\lambda = 0$ is

$$\begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

Eigen vector corresponding $\lambda = 3$.

$(A - 3I)x = 0$.

$$\left(\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix} - \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix} \right) x = 0.$$

$$\begin{bmatrix} 5 & -6 & 2 \\ -6 & 4 & -4 \\ 2 & -4 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0.$$

By Gauss elimination method,

$$\begin{array}{l} R_2 \to 5R_2 + 6R_1 \\ R_3 \to 5R_3 - 2R_1 \end{array} \begin{bmatrix} 5 & -6 & 2 \\ 0 & -16 & -8 \\ 0 & -8 & -4 \end{bmatrix} \underset{\sim}{R_3 \to 2R_3 - R_2} \begin{bmatrix} 5 & -6 & 2 \\ 0 & -16 & -8 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\therefore\ 5x_1 - 6x_2 + 2x_3 = 0$$

$$-16x_2 - 8x_3 = 0$$

$$2x_2 + x_3 = 0$$

$$x_3 = -2x_2.$$

$$5x_1 - 6x_2 + 2x - 2x_2 = 0$$

$$5x_1 - 6x_2 - 4x_2 = 0.$$

$R(A) = 2 < n.$

$$5x_1 - 10x_2 = 0. \Rightarrow x_1 - 2x_2 = 0$$

$\Rightarrow$ Infinitely many soln

$$x_2 = a. \Rightarrow x_1 - 2a = 0$$

$$x_1 = 2a$$

$\therefore$ Eigen vector corresponding $\lambda = 3$ is

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2a \\ a \\ -2a \end{bmatrix} = a\begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}$$

$$\text{Eigen vector} = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}$$

$|||^{rly}$ Eigen vector corresponding $\lambda = 15$ is

$$\begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}$$

12. a) Solve the following linear system of equations using Gauss elimination method.

$$2x - 2y + 4z = 0$$
$$-3x + 3y - 6z + 5w = 15$$
$$x - y + 2z = 0.$$

Soln) $A = \left[\begin{array}{cccc|c} 2 & -2 & 4 & 0 & 0 \\ -3 & 3 & -6 & 5 & 15 \\ 1 & -1 & 2 & 0 & 0 \end{array}\right] \cong \left[\begin{array}{cccc|c} 1 & -1 & 2 & 0 & 0 \\ -3 & 3 & -6 & 5 & 15 \\ 2 & -2 & 4 & 0 & 0 \end{array}\right]$

$R_2 \to R_2 + 3R_1$
$\cong$
$R_3 \to R_3 - 2R_1$

$\left[\begin{array}{cccc|c} 1 & -1 & 2 & 0 & 0 \\ 0 & 0 & 0 & 5 & 15 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$

$R(AB) = 2$
$R(A) = 2$
$R(A) = R(AB) < n = 4.$

$\Rightarrow$ Consistent and solution.

$x = b - 2a, \quad y = b, \quad z = a, \quad w = 3$

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b). Find the matrix of transformation that diagonalize the matrix $A = \begin{bmatrix} -5 & 2 \\ 2 & -2 \end{bmatrix}$ Also write the diagonal matrix soln) characteristic equation is $|A - \lambda I| = 0$

$$\cancel{\Rightarrow 0\lambda^2 + 7\lambda + 6 = 0}$$

$$\cancel{7\lambda + 6 = 0}$$

$$A - \lambda I = \begin{bmatrix} -5 & 2 \\ 2 & -2 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} = \begin{bmatrix} -5-\lambda & 2 \\ 2 & -2-\lambda \end{bmatrix}$$

$$|A - \lambda I| = 0 \Rightarrow \begin{vmatrix} -5-\lambda & 2 \\ 2 & -2-\lambda \end{vmatrix} = 0$$

$$(-5-\lambda)(-2-\lambda) - 4 = 0$$

$$(5+\lambda)(2+\lambda) - 4 = 0$$

$$(5+\lambda)(2+\lambda) = 4.$$

$$10 + 5\lambda + 2\lambda + \lambda^2 = 4$$

$$\lambda^2 + 7\lambda + 10 = 4$$

$$\lambda^2 + 7\lambda + 6 = 0.$$

$$(\lambda+1)(\lambda+6) = 0$$

$$\underline{\lambda = -1, -6.}$$

Eigen vector corresponding $\lambda = -1$ is $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$

Eigenvector corresponding $\lambda = -6$ is $\begin{bmatrix} -2 \\ 1 \end{bmatrix}$

$\therefore$ Modal Matrix $B = \begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix}$

Diagonal matrix, $D = \begin{bmatrix} -1 & 0 \\ 0 & -6 \end{bmatrix}$

Module 2.

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13. a) The length and width of a rectangle are measured with errors of at most 8% and 4% respectively. Use differentials to approximate the maximum percentage error in the calculated area.

Soln) Area $= xy$. $\Rightarrow A = xy$ $\qquad \frac{dA}{A} = ?$

$\frac{dA}{dx} = y$ $\quad \frac{dA}{dy} = x$

$\frac{dx}{x} \times 100 \leq 8$ $\qquad \frac{dy}{y} \times 100 \leq 4$

$dA = A_x dx + A_y dy = y\,dx + x\,dy.$

$$\frac{dA}{A} = \frac{y\,dx}{A} + \frac{x\,dy}{A}.$$

$$\frac{dA}{A} = \frac{y\,dx}{xy} + \frac{x\,dy}{xy} = \frac{dx}{x} + \frac{dy}{y}.$$

$$\frac{dA}{A} \times 100 = \frac{dx}{x} \times 100 + \frac{dy}{y} \times 100$$

$$\leq 3 + 4.$$

$$\leq 7.$$

$$\frac{dA}{A} \leq \frac{7}{100} = 0.07.$$

percentage error $= 7\%$

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b). Find the local linear approximation L of $f(x,y,z) = xyz$ at the point $P(1,2,3)$. Compute the error in approximation f by L at the point $Q(1.001, 2.002, 3.003)$

Soln) $L(x,y,z) = f(a,b,c) + (x-a) f_x(a,b,c) + (y-b) f_y(a,b,c) + f_z(a,b,c)(z-c)$

$$= 6 + (x-1)\,6 + (y-2)\,3 + 2(z-c.$$

$L(1.001, 2.002, 3.003) = 6.018$

$$\text{Error} = \left| f(1.001, 2.002, 3.003) - L(1.001, 2.002, 3.003) \right|$$

$$= .0000180$$

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4 a) If $w = f(P, Q, R)$ where $P = x - y$, $Q = y - z$, $R = z - x$ P.T $\frac{\partial w}{\partial x} + \frac{\partial w}{\partial y} + \frac{\partial w}{\partial z} = 0$.

$$w \to (P, Q, R) \to (x, y, z).$$

$$\frac{\partial w}{\partial x} = \frac{\partial w}{\partial P} \cdot \frac{\partial P}{\partial x} + \frac{\partial w}{\partial Q} \cdot \frac{\partial Q}{\partial x} + \frac{\partial w}{\partial R} \cdot \frac{\partial R}{\partial x}.$$

$$= \frac{\partial w}{\partial P} \cdot 1 + \frac{\partial w}{\partial Q} \times 0 + \frac{\partial w}{\partial R} \cdot (-1).$$

$$= \frac{\partial w}{\partial P} - \frac{\partial w}{\partial R} \quad \text{--- (1)}$$

$$\frac{\partial w}{\partial y} = \frac{\partial w}{\partial P} \cdot \frac{\partial P}{\partial y} + \frac{\partial w}{\partial Q} \cdot \frac{\partial Q}{\partial y} + \frac{\partial w}{\partial R} \cdot \frac{\partial R}{\partial y}.$$

$$= \frac{\partial w}{\partial P}(-1) + \frac{\partial w}{\partial Q} \times 1 + \frac{\partial w}{\partial R} \times 0.$$

$$= -\frac{\partial w}{\partial P} + \frac{\partial w}{\partial Q} \quad \text{--- (2)}$$

$$\frac{\partial w}{\partial Z} = \frac{\partial w}{\partial P}\cdot\frac{\partial P}{\partial z} + \frac{\partial w}{\partial Q}\cdot\frac{\partial Q}{\partial Z} + \frac{\partial w}{\partial R}\cdot\frac{\partial R}{\partial Z}$$

$$= \frac{\partial w}{\partial P}\cdot x\, 0 + \frac{\partial w}{\partial Q} \times (-1) + \frac{\partial w}{\partial R}\cdot 1$$

$$= -\frac{\partial w}{\partial Q} + \frac{\partial w}{\partial R} \quad \text{---} (3)$$

(1) + (2) + (3) $\Rightarrow$

$$\frac{\partial w}{\partial x} + \frac{\partial w}{\partial y} + \frac{\partial w}{\partial Z} = \frac{\partial w}{\partial P} - \frac{\partial w}{\partial R} + -\frac{\partial w}{\partial P} + \frac{\partial w}{\partial Q}$$

$$-\frac{\partial w}{\partial Q} + \frac{\partial w}{\partial R}$$

$$= \underline{\underline{0}}$$

14(b). Locate all relative extrema and saddle points of $f(x,y) = 4xy - x^4 - y^4$

ans)

$f_x = 4y - 4x^3$ $\quad f_{xy} = 4$

$f_y = 4x - 4y^3$ $\quad f_x = 0 \Rightarrow 4y - 4x^3 = 0$

$y - x^3 = 0$

$f_{xx} = -12x^2$ $\quad y = x^3$

$f_{yy} = -12y^2$ $\quad f_y = 0 \Rightarrow 4x - 4y^3 = 0$

$x = y^3$

Solving $y = x^3$ and $x = y^3$ we get $(0,0)$, $(1,1)$, $(-1,-1)$ as solutions. Thus

Critical points are $(0,0)$, $(1,1)$, $(-1,-1)$

$$D = f_{xx} f_{yy} - (f_{xy})^2$$

$$= -12x^2 \cdot -12y^2 - (4)^2$$

$$= 144x^2y^2 - 16$$

<u>At $(0,0)$</u>

$D = -16$

$D < 0$. $\Rightarrow (0,0)$ is a saddle point or f has neither maximum nor minimum at $(0,0)$

<u>At $(1,1)$.</u>

$D = 144 - 16 = 128 > 0$.

f_{xx} at $(1,1) = -12$

Thus $D > 0$, $f_{xx} < 0$

$\Rightarrow f$ attains its maximum value at $(1,1)$

and maximum value $= 2$ $(f(1,1))$

At $(-1,-1)$

$D = 128$

$D > 0$.

$f_{xx} = -12x^2$

f_{xx} at $(-1,-1) = -12$.

Thus $D > 0$; $f_{xx} < 0$.

$\Rightarrow$ f attains its maximum value at $(-1,-1)$ and maximum value $= f(-1,-1) = 2$.

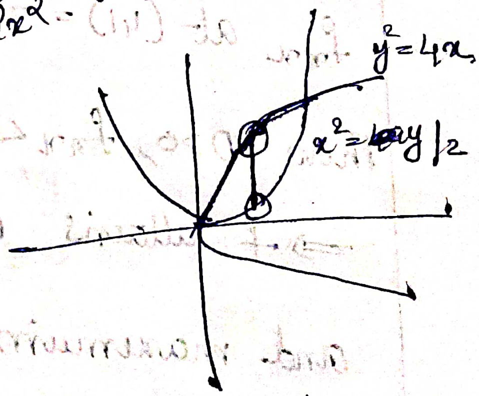
Module 3.

15 a). Find the area bounded by the parabolas $y^2 = 4x$ and $x^2 = y/2$.

Soln)

$$\text{Area} = \iint dA = \int_0^1 \int_{y=2x^2}^{2\sqrt{x}} dy\,dx$$

$$= \int_0^1 \Big[y\Big]_{2x^2}^{2\sqrt{x}} dx$$



$$= \int_0^1 \left(2\sqrt{x} - 2x^2\right) dx$$

$$= \left[2 \cdot \frac{x^{3/2}}{3/2} - 2 \cdot \frac{x^3}{3}\right]_0^1 = \frac{4}{3} - \frac{2}{3} = 2/3$$

15 b) Evaluate $\int_{-2}^{2}\int_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} e^{-(x^2+y^2)}\,dx\,dy$. using polar co-ordinates

Soln) $\int_0^{2\pi}\int_0^{2} e^{-r^2}\, r\,dr\,d\theta$.

$r^2 = u$

$2r\,dr = du$.

$r\,dr = du/2$.

$r=0 \Rightarrow u=0$

$r=2 \Rightarrow u=4$.

$\Rightarrow \int_0^{2\pi}\int_0^{2} e^{-r^2}\, r\,dr\,d\theta$.

$= \int_0^{2\pi} \frac{1}{2}\left[1-e^{-4}\right] d\theta$.

$= \frac{1}{2}\left[1-e^{-4}\right]\left[\theta\right]_0^{2\pi}$

$= \frac{2\pi}{2}\left[1-e^{-4}\right]$

$= \underline{\underline{\pi\left[1-e^{-4}\right]}}$

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16 a). Evaluate $\int_0^1\int_y^1 \frac{x}{x^2+y^2}\,dx\,dy$ by reversing the order of integration.

Soln)
$$\int_0^1 \int_y^1 \frac{x}{x^2+y^2}\, dx\, dy.$$

$$= \int_0^1 \int_0^x \frac{x}{x^2+y^2}\, dy\, dx$$

$$= \int_0^1 \left[x \cdot \frac{1}{x} \tan^{-1}(y/x) \right]_0^x dx$$

$$= \int_0^1 \left[\tan^{-1} y/x\right]_0^x dx.$$

$$= \int_0^1 \left(\tan^{-1} 1 - \tan^{-1} 0\right) dx.$$

$$= \int_0^1 \frac{\pi}{4} - 0 \; dx. \quad = \frac{\pi}{4}(x)_0^1 = \underline{\underline{\pi/4}}$$

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b). Use triple integral to find the volume of the solid within the cylinder $x^2+y^2=9$ b/w the planes $z=1$ and $x+z=5$

Soln). Volume, $V = \iiint dV = \iint_R (4-x)\, dA.$

$$= \int_{-3}^{3} \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} (4-x)\, dy\, dx.$$

$$= \int_{-3}^{3} (4-x) \left[y \right]_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} dx$$

$$= \int_{-3}^{3} (4-x) \left[\sqrt{9-x^2} + \sqrt{9-x^2} \right] dx$$

$$= \int_{-3}^{3} (4-x)\, 2\sqrt{9-x^2}\, dx.$$

$$= \int_{-3}^{3} 4 \times 2 \sqrt{9-x^2}\, dx - 2\int_{-3}^{3} x \sqrt{9-x^2}\, dx.$$

$$= 8 \int_{-3}^{3} \sqrt{9-x^2}\, dx - \cancel{x}\, 0.$$

$$\boxed{\int_{-a}^{a} \sqrt{a^2-x^2} = \frac{\pi a^2}{2}}$$

$$\cancel{= \left[x\sqrt{9-x^2} + 9 \sin^{-1} \frac{x}{3} \right]_{-3}^{3}}$$

$$\cancel{= \left[9 \sin^{-1} 1 - 9 \sin^{-1}(-1) \right]}$$

$9-x^2=u$

$-2x\,dx = du$

$x\,dx = \frac{du}{-2}$

$x=3 \Rightarrow u=0$

$x=-3$

$$= 8 \times \frac{\pi . 3^2}{2} = \frac{8 \times 9 \times \pi}{2} = \underline{\underline{36\pi}}$$

$4 \cancel{8} \times \frac{\pi \times 9}{\cancel{2}}$

17 a) Test the convergence of 1) $\sum_{k=1}^{\infty} \frac{3k^3 - 2k^2 + 4}{k^7 - k^3 + 2}$

2) $\sum_{k=1}^{\infty} \frac{k^k}{k!}$

1) $\sum_{k=1}^{\infty} \frac{3k^3 - 2k^2 + 4}{k^7 - k^3 + 2}$

$a_k = \frac{3k^3 - 2k^2 + 4}{k^7 - k^3 + 2}$. $b_k = \frac{3}{k^4}$. which is a P series

$\lim_{k \to \infty} \frac{a_k}{b_k} = 1$. , By limit comparison test.

Σa_k cgs since Σb_k is cgnt

2) $\sum_{k=1}^{\infty} \frac{k^k}{k!}$

$a_k = \frac{k^k}{k!}$ $a_{k+1} = \frac{(k+1)^{k+1}}{(k+1)!}$

$\lim_{k \to \infty} \frac{a_{k+1}}{a_k} = \lim \frac{(k+1)^{k+1}}{(k+1)!} \times \frac{k!}{k^k} = \lim_{k \to \infty} \frac{(k+1)(k+1)^k}{k^k (k+1)}$.

$= \lim_{k \to \infty} \left(\frac{k+1}{k}\right)^k = e > 1$.

$\therefore$ By ratio test Σa_k dgs.

17 (b) Test whether the series $\sum_{k=1}^{\infty} (-1)^{k+1} \frac{1}{\sqrt{k+1}}$ is absolutely cgnt or conditionally cgnt.

Soln) $a_k = \frac{1}{\sqrt{k+1}}$

$\lim_{k\to\infty} a_k = 0$

$\frac{a_{k+1}}{a_k} \leq 1 \quad \forall k.$

Then by alternating series test $\sum a_k (-1)^{k+1}$ cgs

$u_k = (-1)^{k+1} \frac{1}{\sqrt{k+1}} \Rightarrow |u_k| = \frac{1}{\sqrt{k+1}}$.

Let $v_k = \frac{1}{\sqrt{k}}$.

Then by limit comparison test, $\sum |u_k|$ dgs

$\Rightarrow \sum u_k$ dgs absolutely $\Rightarrow$ Conditionally cgnt.

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18 Q. Test the convergence of the series

$\frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \frac{1}{4.5} + \cdots$

Soln) $\frac{1}{1\cdot 2}+\frac{1}{2\cdot 3}+\cdots = \sum_{k=1}^{\infty}\frac{1}{k(k+1)}$.

$a_k = \frac{1}{k(k+1)}$. Let $b_k = \frac{1}{k^2}$.

$\lim_{k\to\infty}\frac{a_k}{b_k} = 1$.

Then by limit comparison test $\sum a_k$ cgs

Since $\sum b_k$ is a cgnt p series $(p=2)$

8(b). Test the Convergence of 1) $\sum_{k=1}^{\infty}\left(\frac{k}{k+1}\right)^{k^2}$. 2) $\sum_{k=1}^{\infty}\frac{T^k}{k!}$

Soln) 1) $a_k = \left(\frac{k}{k+1}\right)^{k^2}$.

$\lim_{k\to\infty}(a_k)^{1/k} = \frac{1}{e} < 1$.

Then by root test $\sum a_k$ cgs.

2) $a_k = \frac{T^k}{k!}$, $a_{k+1} = \frac{T^{k+1}}{(k+1)!}$

$\lim_{k\to\infty}\frac{a_{k+1}}{a_k} = 0 < 1$.

Then by ratio test $\sum a_k$ cgs.

19 a) Find the fourier series expansion of $f(x) = x - x^2$ in the range $(-1, 1)$

Soln. $l = \frac{1-1}{2} = 1$

$$a_0 = \frac{1}{l}\int_{-1}^{1} f(x)\,dx = \frac{1}{1}\int_{-1}^{1} x - x^2\,dx$$

$$= \left[\frac{x^2}{2} - \frac{x^3}{3}\right]_{-1}^{1}$$

$$= \left(\frac{1}{2} - \frac{1}{3}\right) - \left(\frac{1}{2} - \frac{-1}{3}\right)$$

$$= \frac{1}{6} - \frac{5}{6} = \frac{-4}{6} = \frac{-2}{3}$$

$$a_n = \frac{1}{l}\int_{-1}^{1} f(x)\cos\left(\frac{n\pi x}{l}\right) dx.$$

$$= \frac{1}{1}\int_{-1}^{1}(x - x^2)\cdot \cos n\pi x\,dx.$$

$$= \int_{-1}^{1}(x - x^2)\cos n\pi x\,dx.$$

$$= \int_{-1}^{1} x\cos n\pi x\,dx - \int_{-1}^{1} x^2\cos n\pi x\,dx \quad \text{--- (1)}$$

$$\int_{-1}^{1} x \cos n\pi x \, dx = \left[x \cdot \frac{\sin n\pi x}{n\pi} - \frac{-\cos n\pi x}{n^2\pi^2}\right]_{-1}^{1}$$

$$= \left[x \frac{\sin n\pi x}{n\pi} + \frac{\cos n\pi x}{n^2\pi^2}\right]_{-1}^{1}$$

$$= \left(\frac{\sin n\pi}{n\pi} + \frac{\cos n\pi}{n^2\pi^2}\right) - \left(+1 \cdot \frac{\sin n\pi}{n\pi} + \frac{\cos n\pi}{n^2\pi^2}\right)$$

$$= \frac{(-1)^n}{n^2\pi^2} - \frac{(-1)^n}{n^2\pi^2} = 0.$$

$$\int x^2 \cos n\pi x \, dx = \left[x^2 \frac{\sin n\pi x}{n\pi} - 2x \frac{-\cancel{\cos} n\pi x}{n^2\pi^2} \cancel{+} 2\frac{\cancel{\sin} n\pi x}{n^3\pi^3}\right]_{-1}$$

$$= \left[x^2 \frac{\sin n\pi x}{n\pi} + 2x \frac{\overset{\cos}{\cancel{\sin}}\, n\pi x}{n^2\pi^2} \cancel{+} 2\frac{\overset{\sin}{\cancel{\cos}}\, n\pi x}{n^3\pi^3}\right]_{-1}^{1}$$

$$= \frac{2(-1)^n}{n^2\pi^2} - \left(\frac{2x - 1 \cos n\pi}{n^2\pi^2}\right)$$

$$= \frac{2 \cdot (-1)^n + 2(-1)^n}{n^2\pi^2} = \frac{4(-1)^n}{n^2\pi^2}.$$

$$(P) \Rightarrow \quad 0 - \frac{4(-1)^n}{n^2\pi^2} = \frac{4(-1)^{n+1}}{n^2\pi^2}$$

$$b_n = \frac{1}{l}\int_{-l}^{l} f(x)\, \sin\left(\frac{n\pi x}{l}\right) dx$$

$$= \frac{1}{1}\int_{-1}^{1} (x - x^2)\, \sin n\pi x\, dx$$

$$= \int_{-1}^{1} x \sin n\pi x\, dx - \int_{-1}^{1} x^2 \sin n\pi x\, dx \quad \text{—— (1)}$$

$$\int_{-1}^{1} x \sin n\pi x\, dx = \left[x \frac{-\cos n\pi x}{n\pi} - \frac{-\sin n\pi x}{n^2\pi^2} \right]_{-1}^{1}$$

$$= \frac{-1(-1)^n}{n\pi} - \frac{(-1)\,^{-}(-1)^n}{n\pi}$$

$$= \frac{(-1)^{n+1}}{n\pi} + \frac{(-1)^{n+1}}{n\pi} = \frac{2(-1)^{n+1}}{n\pi}$$

$$\int x^2 \sin n\pi x\, dx = 0.$$

$$(1) \Rightarrow b_n = \frac{2(-1)^{n+1}}{n\pi}$$

$$\therefore f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right)$$

$$= \frac{-1}{3} + \sum_{n=1}^{\infty} \frac{4(-1)^{n+1}}{n^2\pi^2} \cos(n\pi x) + \sum_{n=1}^{\infty} \frac{2(-1)^{n+1}}{n\pi} \sin n$$

19(b) Obtain the half range Fourier cosine series of $f(x) = e^{-x}$ in $0<x<2$.

$$a_0 = \frac{2}{l}\int_0^l f(x)\,dx.$$

$$l = 2-0 = 2.$$

$$a_0 = \frac{2}{2}\int_0^2 e^{-x}\,dx = \left[\frac{e^{-x}}{-1}\right]_0^2 = -\left[e^{-2}-1\right]$$

$$= \underline{\underline{1-e^{-2}}}$$

$$a_n = \frac{2}{l}\int_0^l f(x)\cos\left(\frac{n\pi x}{l}\right)dx$$

$$= \frac{2}{2}\int_0^2 f(x)\cos\left(\frac{n\pi x}{2}\right)dx.$$

$$= \int_0^2 e^{-x}\cos\left(\frac{n\pi x}{2}\right)dx.$$

$$a_n = \frac{1-e^{-2}\cos n\pi}{1+\left(\frac{n\pi}{2}\right)^2}$$

$$\therefore f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right)$$

$$= \frac{1-e^{-2}}{2} + \sum_{n=1}^{\infty} \frac{1-e^{-2}\cos n\pi}{1+\left(\frac{n\pi}{2}\right)^2}\cos\left(\frac{n\pi x}{2}\right)$$

20 a) Find the Fourier series expansion of $f(x) = x^2$ in the interval $-\pi < x < \pi$ Hence S.T $1 - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \cdots = \frac{\pi^2}{12}$.

$$a_0 = \frac{2\pi^2}{3}$$

$$\frac{\pi - -\pi}{2}$$

$$l = \pi$$

$$a_n = \frac{4}{n^2}(-1)^n$$

$$b_n = 0.$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right)$$

$$= \frac{2\pi^2}{3 \cdot 2} + \sum_{n=1}^{\infty} a_n \cos nx + 0.$$

$$= \frac{\pi^2}{3} + a_1 \cos x + a_2 \cos 2x + a_3 \cos 3x + \cdots$$

$x = 0$

$$0 = \frac{\pi^2}{3} + a_1 \cos 0 + a_2 \cos 0 + a_3 \cos 0 + \cdots$$

$$= \frac{\pi^2}{3} + \frac{4}{1^2}(-1)^1 + \frac{4}{2^2}(-1)^2 + \frac{4(-1)^3}{3^2}$$

$$0 = \frac{\pi^2}{3} - 4 + \frac{4}{2^2} - \frac{4}{3^2} + \cdots$$

$$\Rightarrow \frac{-\pi^2}{3} = -4\left[1-\frac{1}{2^2}+\frac{1}{3^2}-\cdots\right]$$

$$\cdot \frac{\pi^2}{12} = 1-\frac{1}{2^2}+\frac{1}{3^2}+\cdots$$

20(b). obtain the half range Fourier sine series of

$$f(x) = \begin{cases} x & 0<x<\pi/2 \\ \pi-x & \frac{\pi}{2}<x<\pi \end{cases}$$

$l = \pi - 0$
$= \pi$

Soln) $$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right) = \sum_{n=1}^{\infty} b_n \sin nx$$

$$b_n = \frac{2}{l}\int_0^l f(x)\sin\left(\frac{n\pi x}{l}\right)dx$$

$l = \pi$

$$b_n = \frac{2}{\pi}\left[\int_0^{\pi/2} x\sin nx\,dx + \int_{\pi/2}^{\pi}(\pi-x)\sin nx\,dx\right]$$

$$b_n = \frac{4}{n^2\pi}\sin\left(\frac{n\pi}{2}\right)$$

$$\therefore f(x) = \sum_{n=1}^{\infty}\frac{4}{n^2\pi}\sin\left(\frac{n\pi}{2}\right)\sin nx$$